

DermaForce: RAG + QLoRA for Hebrew Dermatology Board-Exam QA

Applied Language Models — Group Project B (solo submission)

1. Introduction & motivation

We extend a working Hebrew-to-English RAG chatbot over *Bologna's Dermatology* (5th ed.) into a measured research system, and evaluate it against a source of ground truth most course RAG projects don't have: **949 real questions from seven sittings (2021–2026) of the Israeli Medical Association's Stage-A dermatology board exam**, each with a human expert's page/chapter citation into the same textbook already indexed. This closes the main methodological gap of a synthetic-only evaluation (a model grading its own generated questions) and adds the assignment's two hard requirements a pure product build doesn't need: a measured retrieval ablation and a QLoRA fine-tune.

The exam data also exposes two real, quantifiable ceilings for a text-only RAG system: **image-dependent questions** (17% of the real corpus — a generator-side ceiling) and **table/figure-pointer references** (a retriever-side ceiling, since captions are dropped at extraction time). Both are measured below rather than treated as excuses.

2. Datasets & preprocessing

	Bologna corpus + synthetic QA	Israeli board-exam corpus
Source	<i>Bologna Dermatology</i> 5th ed., 3,204 pages	7 sittings, 2021–2026, IMA Stage-A
Size	6,115 chunks (400w/80-overlap) → 1,219 synthetic QA items	949/950 MCQs parsed
Ground truth	Claude-generated, gold = exact source chunk	Human expert page/chapter citation — independent
Split	1,000 train / 120 val / 99 test, by chunk (zero leakage)	Temporal: develop 2021–2024 (n=517 text-answerable) / hold out 2025+2026 (n=258 usable)
Licensing	Elsevier — gitignored, small sample committed	IMA — gitignored, small sample committed

Parsing the exam corpus required per-file format detection across the seven years: two numbering conventions (N. vs. a reversed .N), two distinct broken-glyph bugs, three reference-file structures (tabular page-only / prose book+page / prose book+chapter+page), one cross-version alignment table (2023), and crop-box intersection (`not get_images()`) for image-page association, since one year's PDF shares its content stream across all 44 pages. `pipeline/02_chunk.py` was also fixed: the original chunker accumulated whole pages until crossing a word target, but a Bologna page is ~700 words, so the window always overshot on page 1 and closed immediately — median 766 words/chunk with **zero effective overlap**. The rewritten word-stream chunker produces an exact 400-word median with exact 80-word overlap on every pair.

3. Model design

```
Hebrew MCQ --> glyph fix --> translate to EN --> retrieve (EN corpus) --> score 4 options
PRODUCT: Qdrant + OpenAI embeds → Claude → answer → verifier badge
MEASUREMENT: FAISS index variants → retrieval metrics
Claude-distilled QA → QLoRA Qwen2.5-3B-Instruct
```

Four design decisions: (1) FAISS for measurement (no infra to reproduce), Qdrant stays for the product; (2) translate the *query*, not the 2,521-chunk corpus — three orders of magnitude cheaper; (3) score MCQ options by length-normalised log-likelihood, not by generating a letter — the standard way to make a no-RAG base-model arm comparable; (4) `backend/llm.py` behind a Generator protocol (`ClaudeGenerator` / `LoRAGenerator`), so the baseline-vs-fine-tuned comparison is a live, demoable flip in the product UI.

Four evaluation arms, on the 258 held-out 2025+2026 questions: **A.** Qwen2.5-3B base, no RAG (closed-book floor) · **B.** base + RAG (retrieval alone) · **C.** LoRA + RAG (the fine-tuning deliverable) · **D.** Claude + RAG (upper bound / teacher — cannot use log-likelihood scoring since Anthropic exposes no logprobs, so it generates the Hebrew letter directly reading the un-translated stem: a documented, deliberate scoring asymmetry).

4. Training & metrics

Retrieval (eval/eval_retrieval.py): 3 chunk configs × 2 embedders (bge-small-en-v1.5 free, text-embedding-3-large paid) × 3 query-language variants × reranker on/off, scored as Hit@{1,3,5,10} and MRR@10 against two independent gold sources (synthetic exact-chunk, expert page/chapter citation).

QLoRA fine-tune (training/): Qwen2.5-3B-Instruct, 4-bit NF4, LoRA rank $\in \{8, 16, 32\}$ ($\alpha=2r$, dropout 0.05, lr $2e-4$, 2 epochs), trained on Colab. Target: citation-grounded answers in the product's own [Chapter, p.N] format, distilled from Claude, with training examples where the actual retriever misses the gold chunk relabeled to an explicit abstention — so the model isn't taught to state facts absent from its own shown context.

Generation eval: MCQ accuracy (likelihood-scored), macro-F1 (position bias), Hit@k/MRR against expert references, abstention rate on 24 genuinely-out-of-corpus questions, ROUGE-L/BLEU/citation-accuracy/LLM-judge groundedness on the synthetic test slice.

5. Results

4-arm MCQ accuracy (n=258; random floor 0.25; ceiling = 86.3% given 47/258 questions are image-dependent and unanswerable from text):

Arm	Accuracy	Text-answerable	Image-dependent	% of ceiling
A. base, no RAG	0.225	0.218	0.255	26.0%
B. base + RAG	0.302	0.308	0.277	35.0%
C. LoRA r16 + RAG	0.318	0.313	0.340	36.8%
D. Claude + RAG	0.721	0.754	0.574	83.5%

Retrieval alone buys arm B +7.7pp over the closed-book floor; the LoRA fine-tune buys a further +1.6pp — real but modest, consistent with r=32's measured overfitting (see below) and with a shallower training-time retrieval depth (2–3 chunks at a 2048-token budget) than the deployed product's top-8. Claude (D) sits 40 points above C, and framed against the ceiling — 83.5% of everything a text-only system could get right, not "72% accurate" — the gap is concentrated in translation/retrieval noise and reasoning depth, not architecture.

Cross-lingual retrieval — the headline experiment: translating the Hebrew query to English before retrieval buys bge-small (English-only) **+19% relative Hit@10** (0.106→0.124→0.128 across raw/translated/+glossary); text-embedding-3-large (real cross-lingual capability) stays flat (0.114→0.112→0.116). Two hand-read cases confirm the mechanism: a raw-Hebrew query for an atopic-dermatitis question retrieves top-1 from *Fungal Diseases* (unrelated); translated, it retrieves top-1 from *Atopic Dermatitis* — the English-only embedder has no lexical grip on untranslated Hebrew, and translation restores literal term overlap it can use directly.

Reranking (ms-marco-MiniLM-L-6-v2) improved MRR@10 in **every one of 6 query-set × embedder combinations tested** — a clean, low-risk addition. **Table/figure-pointer references** (37/517 dev questions) retrieve **3.7–4.6× worse** than ordinary page references (Hit@1 = 0% in both query languages) — the direct cost of dropping figure captions at extraction time, a distinct, retriever-side failure mode from the generator-side image-dependent-question ceiling.

Position bias: the exam's own answer distribution is skewed (option A correct 75/258 times, D only 42 — a real exam-writing convention). Arm A shows genuine *model* bias on top of that, over-predicting the last option 2× its true rate; arm D tracks the true distribution almost exactly (macro-F1 ≈ accuracy), consistent with larger models showing less MCQ selection bias.

QLoRA loss curves: val loss plateaus flat (~0.29–0.31) for all three ranks from step ~50; train loss keeps dropping fastest for $r=32$ — the train/val gap widening monotonically with rank is a measured overfitting signal, not a prediction. $r=16$ was carried into the 4-arm comparison as the rank least likely to be overfit while retaining real capacity.

Two measurement artifacts caught before being reported as findings: multi-hop references score *better* on Hit@1/MRR (0.087 vs. 0.028) only because they resolve to a $1.7\times$ larger gold-chunk set on average — not because retrieval reasons across locations better; and 97.3% of exam-dev questions overlap the synthetic set's sampled content (stratified sampling already touches all 160 chapters), which rules out "coverage gap" as the explanation for the synthetic-vs-exam accuracy gap and leaves the already-documented one standing: exam citations point at *where a topic is discussed*, not an exact paragraph.

6. Limitations

Translation is a real, now-demonstrated (not just aggregate) error source, though translation calls themselves rarely fail outright ($\leq 1/500$ – $1/200$ across every batch run). The printed-to-PDF page map is an interpolation (122/160 chapters anchored, rest interpolated; validated to 1–2 pages), so every non-chapter-tagged reference inherits residual uncertainty. Arm D's letter-generation scoring is not directly comparable to A/B/C's log-likelihood scoring. Both corpora are copyrighted and not redistributed. Image-dependent questions are a hard architectural ceiling for any text-only system, quantified rather than assumed.

7. Conclusion & future work

A real RAG product, measured honestly against independent human ground truth rather than its own generated questions, clears the assignment's retrieval-ablation and fine-tuning requirements with a full 4-arm comparison and a verifier agent that demonstrably distinguishes grounded claims from fabricated citations. The largest open levers, in order of expected payoff: re-including figure captions as retrievable text (raises the measured ceiling rather than just describing it), a deeper training-time retrieval budget for the LoRA arm (closer to the product's real top-8), and a larger/better-curated multi-hop and table/figure evaluation slice now that both are cleanly isolated as distinct failure classes.